

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			Carbon dioxide concentration increased by approximately 10ppm every 10 years ☐ Carbon dioxide concentration increased more between 1970 and 2010 than it did between 1930 and 1960 ☒ Carbon dioxide concentration increased more between 1930 and 1960 than it did between 1970 and 2010 ☐ There is no trend to the change in carbon dioxide concentration between 1930 and 2010 ☐			1	1	1	
	(b)			as carbon dioxide concentration increases (from 1930 to 2010) then so does the Earth's mean temperature (so good evidence) to support this claim (1) award (1) for any of following / OWTTE - as solar activity increases from 1930 to 1960 then so does Earth's mean temperature but after 1960 solar activity decreases but Earth's average temperature continues to increase (so not good evidence) - solar activity does not change much between 1930 and 2010 but the temperature increases (so not good evidence) - solar activity decreasing at the end but temperature still increasing (so not good evidence) if neither of these marks credited award (1) for 'data suggests that claim is true for carbon dioxide but not true for solar activity' / OWTTE			2	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			award (1) for any of following - this is only one piece of evidence / not enough evidence to prove global dimming / more evidence needed from other events - this is only local information (for a 20-mile radius) around Mount Pinatubo - more information is needed about the temperature values before 1990			1	1		
	(d)			carbon capture / carbon capture and storage	1			1		
	(e)			green plants evolved which photosynthesised (thus absorbing carbon dioxide) (1) award (1) each for any two of following (some) carbon dioxide became locked into fossil fuels (some) carbon dioxide became locked into shells of marine animals / limestone / chalk / rocks (some) carbon dioxide was absorbed by oceans / seas	3			3		
	(f)			Ca(OH) ₂		1		1		
				Question 7 total	4	1	4	9	1	0